order16 RESULTS
PREREG sha256 d0cafdcf63675c31801997183452a27a9557e1082e1ec8a29b52da69a2c6c51e (written before numerics).

Category A
- A. Line terms closed form: W − p = ½(1+s1)h2 + ½(1+s2)h1 + h1h2 − ½g1 − ½g3, finite supports; five line types (half-lines along e1/e2 → partial thetas with partners Σw e(w²τ), Σm e(7m²τ/2); points; full lines ν1=c → weight-½ theta in 7m²/2; full lines n+21k=M → weight-½ theta of norm 595/2). No new weight-3/2 unary theta. Proof in PREREG and paper Prop. prop:lines.
- B. RG1′ necessity: PROVED. Lemma: Σ_{m∈δ+Z} m e(Am²τ+cm) ≡ 0 iff 2δ, 2c, 2cδ ∈ Z. Holomorphic×antiholomorphic product ≡ 0 iff a factor ≡ 0. So the c1-endpoint term vanishes iff ρ_m or ρ_u preserves the companion (RG1′ is iff).
- Map RG2: corollary of the Lemma (see NOTES for an imprecision).

Category B (preregistered checks)
- A1 identity on box 100, 140 companions: 0 mismatches — PASS (weak: an exact identity, the check guards the code only).
- A3: 6 rows ≤ 2.3e−26 (threshold 1e−18) — PASS.
- B1 criterion vs stored zero sets (order14 105 companions, order15 family H, order14 K1): all agree — PASS.

Paper
RTA_FOUNDATION_BOUND_v4.pdf, 128 pages (v3: 123), new section "Shadow structure of the companion series".

order16 NOTES (disclosures)
- Map RG2 as stated in PREREG is imprecise: the P-odd c1-end piece is a product with θ_{7/2}, so it can also vanish if θ_{7/2} ≡ 0, not only on the ρ_u set. The correct statement is "vanish together on the ρ_u set, and the P-odd piece also wherever the weight-½ partner vanishes". No test depended on RG2.
- A1 and A3 are Category B code checks of exact identities; they are not evidence beyond guarding implementation.
- B1 compares against zero sets computed in order14/15, i.e. not independent of those runs' numerics.
- No new Residual–Resonant map was proposed. Maps E1–E3 remain OPEN.
- The "residual geometric bridge" phrase and Arvin's P-odd local / P-even non-local framing remain opinion; the paper uses only a descriptive remark.
- Paper v4: Prop. prop:M2 is computer-assisted exact (embedding certificate); the identification of Zwegers' law with (35^{-1/2}M, T) and 𝔸(Φ_orth)=S^E remain Category B (App. D items sh1–sh4).
- Runs: nice, OMP_NUM_THREADS=1, <3 GB. Nothing pushed or sent.